

AI Says I'm Better: Evaluating the Effect of AI Defer on Users. A Study Protocol

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Abstract. The integration of AI into decision support systems raises concerns about overreliance and distrust. To address this, we propose an experimental protocol combining Learning to Defer (LtD)—where AI delegates decisions to humans when appropriate—and Explainable AI (XAI), which provides users with decision rationales. Our study investigates how these approaches impact human decision-making, particularly in high-stakes contexts. Participants will classify noisy images from ImageNet under three between-subjects conditions: Defer (AI defers to user), Defer + XAI (AI provides an explanation), and Hidden Delegation (AI involvement is concealed). Each condition will be tested in neutral and high-stakes scenarios, the latter framed through narratives emphasizing the danger of misclassification. We will assess decision accuracy and reaction times, as well as psychological measures that explore the influence of individual differences (i.e., intolerance to uncertainty and cognitive styles), and emotions (e.g., emotion regulation, and AI-related anxiety). We hypothesize that Defer may prompt more analytical thinking, improving accuracy over Hidden Delegation, while Defer + XAI may further enhance performance. In contrast, Hidden Delegation could promote reliance on intuitive processing. We expect higher accuracy and longer response times in high-stakes conditions. Findings will inform the design of human-AI systems that optimize user engagement and reliability, particularly in domains like clinical decision-making.

Keywords. Human-AI collaboration, Learning to Defer (LtD), Explainable AI (XAI), Decision-making, Decision Support System (DSS), High-stakes scenarios, Emotions, Individual difference, Cognitive Style.

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1. Introduction

The integration of Artificial Intelligence (AI) into human decision-making processes has raised important questions about how users interact with intelligent systems. Especially in high-stakes situations [1], where decisions carry significant consequences, users often oscillate between two opposing tendencies: overreliance, where they shift most of the responsibility to the machine and passively accept its responses, and distrust, which implies low adoption and opposition to the AI tool. Achieving a balance between these extremes is crucial to improve both decision quality and the user's perception of being in control, while the promotion of this balance can foster more effective and trustworthy human-AI collaboration.

From a psychological perspective, effective AI development should not aim to replicate human cognition, but rather to build systems that transparently complement human strengths while acknowledging both human and machine limitations in understanding and creating meaning [2]. Moreover, it is essential to keep the human (both the decision maker and the person that is subject to that decision, e.g., the patient) at the center of the scenario [3]. Because of this, it is essential to experimentally assess whether AI interventions truly support human decision-making and, ultimately, if they have a positive impact on cognitive processes. A central framework in understanding these interactions is the dual-process theory, which distinguishes between intuitive, fast responses (System 1) and more deliberate, analytical reasoning (System 2) [4]. This model supports the understanding of how overreliance can come into place, with repetitive or routine decisions that often lead to increased delegation and automaticity, reinforcing reliance on external responses. It remains, however, an open question in which contexts this tendency is attenuated and when cognitive engagement is maintained. Two AI design strategies aimed at addressing this challenge are *Learning to Defer (LtD)* and *Explainable AI (XAI)*. LtD is a paradigm [5], in which the AI system recognizes its limitations [6] and delegates the decision to a human when the human is more likely to be correct than the AI system, with the explicit aim of achieving the best predictions from the combination of AI and the human expert [7]. This approach appears to be particularly valuable, according to the comprehensive meta-analysis of Vaccaro et al. (2024), which weighed the potential benefits of human-AI collaboration. In particular, their conclusions show that when human initially outperforms the AI, their combination tends to improve the overall performance, while the opposite situation does not reach positive outcomes [8]. Technically, in LtD the model learns to defer adaptively by estimating, for each case, whether the human decision-maker is likely to make a better judgment than the model itself, based on features of the input and past performance, and uses this to decide whether to predict or pass the decision along [4].

Additionally, XAI techniques are designed to provide users with understandable explanations of the AI's outputs, allowing individuals to make more informed decisions while taking into account how humans usually arrive at a response [9]. If the model is not inherently transparent or "white-box", as in the case of decision trees or other interpretable algorithms, researchers can apply post hoc XAI techniques to approximate how the opaque model arrived at a specific decision [10]. In this sense, both LtD and XAI aim to reactivate analytical reasoning by encouraging users to reflect more deeply on the decision-making process. It is therefore essential to experimentally evaluate whether the interventions applied in AI scenarios actually improve human decision-making and positively influence cognitive processes.

Moreover, despite increasing attention to cognitive and contextual factors in human-AI collaboration, individual psychological differences are often underexplored in the existing literature on the topic. Yet, individual variability plays a crucial role in how users interpret and respond to AI inputs. These differences may be cognitive, such as *decision-making styles* [11] (i.e., habitual patterns in how individuals approach and resolve decisions) and *attitudes toward AI* [12] (i.e., beliefs and expectations about AI technologies), or affective, as *intolerance of uncertainty* [13] (i.e., the tendency to find ambiguous or unpredictable situations distressing), and *emotion regulation strategies* [14] (i.e., methods used by individuals to influence their emotional). Neglecting these dimensions may lead to an oversimplified view of human-AI interaction, potentially undermining the development of adaptive, user-centered systems.

A key question is thus: *In which circumstances do humans achieve higher performance, and why?* Understanding this involves examining overreliance as a cognitive construct related to System 1 dominance, while distrust may be influenced by other variables, such as the absence of explanation, but also individual differences. To evaluate the combined effectiveness of LtD and XAI and understand how to align AI systems with human cognitive processes [15] we therefore propose a study protocol that examines the impact of these approaches on human decision-making, particularly in high-stakes contexts, evaluating the conditions under which these techniques successfully trigger analytical reasoning, acting as a “cognitive honk” that brings the human decision-maker back “into the driver’s seat”.

2. Methods

In this study protocol, we propose an experimental setup that revolves around a decision task in which the participants will be asked to correctly classify visual stimuli that will be manipulated with a layer of visual noise.

Dataset. As stimuli, we will consider the noisy version of ImageNet1k introduced by Kerrigan et al. [16]. The dataset consists of 1200 perturbed images representing 16 different categories (clock, bottle, cat, elephant, chair, knife, oven, bird, keyboard, airplane, truck, car, bear, dog, bicycle, boat). Each image is paired with a prediction provided by a human expert who was asked to correctly classify the image.

Deferring system. A deferring system is an ML model that determines whether a certain prediction must be provided by the ML model itself or by an external agent, typically a human expert. For this purpose, we will consider a DenseNet121 architecture with $K+1$ heads, where K is the number of classes of the classification task, e.g., in our case $K = 16$. The K -th head estimates the probability the image belongs to the K -th class, while the $K+1$ -th head models whether the human expert correctly predicts the image class. The difference between the $K+1$ -th head and the maximum of the remaining K heads is called *reject score*. Intuitively, the higher the reject score, the better to defer a certain image to the human expert.

Experiment Setup. We will randomly split the noisy ImageNet1k into training and test sets, using an 80%-20% proportion. We will train the deferring system using the Realizable Surrogate loss proposed by Mozannar et al. [17], which achieves state-of-the-art results on Imagenet1k. We will then compute reject scores on the test set and select

the 20 images with the highest reject score, as done by Hemmer et al, 2023 [18], plus 3 images randomly selected.

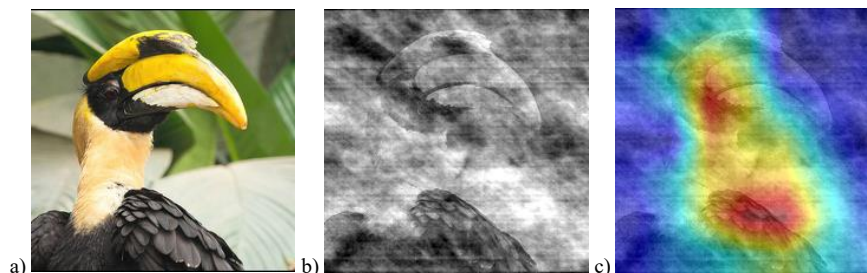


Figure 1. Example of image stimulus (bird): a) original from imagenet-1k, b) image with noise, c) image with noise and visual explanation (XAI)

Participant Task. Following the setup by Hemmer and colleagues [18], the participants will be asked to recognize the 3 randomly chosen images for familiarization and the remaining 20 for the main experiment. We will identify three different between-subjects experimental conditions:

- **LtD:** The system explicitly delegates the decision to the user (e.g., Figure 1b).
- **LtD + XAI:** The system adds an explanation for its decision (e.g., Figure 1c) through a GradCAM.
- **Hidden Delegation:** The participants are not informed that an AI has already processed the image (e.g., Figure 1b).

To increase participant engagement and simulate a high-stakes decision scenario with a high degree of perceived importance, each condition will be presented both with an emotionally engaging narrative and with a neutral one, for control. The narrative will describe the images as containing neutral or dangerous objects, emphasizing that failing to recognize dangerous items could lead to severe consequences. This manipulation is intended to examine whether contextual emotional framing influences decision-making performance and approach.

System 1 and System 2 activation. Participants will be asked to reply to the question “What was your approach to completing the task?” through a Visual Assessment Scale (VAS) ranging from 0 to 100, where 0 = “I relied completely on gut feeling/it was completely intuitive” and 100 = “I relied completely on logical analysis/it was completely deliberate reasoning”. This measure will be used to assess the subjective reasoning style adopted after each image classification. As we assume that the LtD and XAI + LtD conditions should elicit more deliberate reasoning (i.e., System 2 activation), the VAS score will serve as a proxy for the cognitive processing style engaged. Before using this measure in our analysis, we will perform a convergent validity check to ensure it adequately captures the intended construct.

The global task performance will be evaluated based on the accuracy of responses, while response time will be recorded as an additional measure, potentially correlating with the activation of System 2 processing [19].

Dispositional traits influence. Exploratory measures will include participant profiling based on individual dispositional traits that may influence decision-making and interactions with AI-based systems. Specifically, we will assess intolerance to uncertainty, decision-making styles, emotional regulation strategies, and attitudes towards AI. Specifically, we will administer the Intolerance of Uncertainty Scale – Short Form (IUS-12) [20] to measure participants’ tendency to find uncertain situations distressing. To evaluate dispositions toward AI, we will use the General Attitudes towards Artificial Intelligence Scale [21], which assesses both positive and negative beliefs about AI. Emotional regulation strategies will be assessed using the Emotion Regulation Questionnaire (ERQ) [22], focusing on individual differences in emotion regulation strategies (i.e., cognitive reappraisal- the tendency to reinterpret situations in order to alter their emotional impact – vs expressive suppression – the tendency to inhibit the expression of emotions). Finally, the general decision-making style (GDMS) questionnaire [23] will be included to assess participants’ typical approaches to making decisions.

The estimated sample size is approximately 550 participants.

3. Conclusions

We hypothesize that the LtD condition might outperform Hidden Delegation by encouraging users to shift from automatic (System 1) to more deliberate (System 2) processing, as users are made aware that they are taking responsibility for a decision previously considered by the system. The Defer + XAI condition may further enhance this effect by providing users with a visual explanation that supports cognitive engagement and increases their confidence in understanding the system’s reasoning. Conversely, in the Hidden Delegation condition, the lack of contextual cues and responsibility awareness might lead to disengagement or habituation, increasing the likelihood of errors due to reliance on default, intuitive processing. Moreover, the inclusion of emotionally salient scenarios is expected to amplify cognitive engagement, particularly in conditions where users perceive the consequences of errors as meaningful. In such cases, response times might be longer, reflecting more thoughtful processing, and accuracy may improve due to increased attentional resources and self-monitoring.

We also expect that individual differences will play a critical role in modulating these effects. Specifically, individuals with a more analytical decision-making style or positive attitudes toward AI may be more likely to engage thoughtfully with the system, particularly in the LtD + XAI condition, leading to higher accuracy and longer reaction times indicative of System 2 processing. Conversely, those with high intolerance of uncertainty or who consistently rely on suppression as an emotion regulation strategy may show reduced trust or avoidant behavior, especially when explanations are absent, potentially leading to less effective decisions under pressure.

In addition to this, the inclusion of both neutral and emotionally charged narratives allows us to test not only how participants respond to different forms of AI support, but also how contextual emotional salience affects cognitive engagement. This framing is designed to mimic real-world decision-making conditions, where outcomes have tangible consequences, thereby increasing ecological validity. By comparing

performance across emotionally neutral and emotionally evocative conditions, we aim to understand whether these individual traits are stable predictors of behavior or whether their impact is amplified or attenuated by emotional context.

The ultimate goal of this work is indeed to show that users' psychological and emotional diversity matters, not just in general, but in the way it interacts with system design (e.g., explanations, delegation) and context (e.g., stakes). Future AI systems that adapt to user profiles and situational contexts could be more trustworthy, effective, and inclusive. Understanding when and why humans benefit from taking back control in decision-making and how AI systems can support rather than replace that process may lead to more responsible, effective, and trustworthy human-AI collaborations. Moreover, immersive technologies such as virtual reality could be leveraged in future studies to simulate high-stakes environments more vividly [24,25]. This would enhance ecological validity and allow for a more thorough investigation of how users engage with AI support under realistic and emotionally salient conditions.

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